

How do FAIR data principles contribute to developing ethical and resilient infrastructure for AI research across scientific and industrial domains?

Through technical implementations and governance frameworks, FAIR principles create AI research infrastructures that balance ethical responsibility with technical resilience across scientific and industrial applications.

Abstract

FAIR data principles—making data findable, accessible, interoperable, and reusable—serve as a technical and ethical foundation for AI research infrastructures across scientific and industrial domains. In ten studies reviewed, seven report full implementations that employ persistent identifiers, rich metadata catalogues, open APIs, containerized environments (e.g., Docker, Apptainer), and federated or distributed architectures. These technical choices underpin ethical practices such as strict adherence to data protection laws (for example, the General Data Protection Regulation), privacy-preserving computation (through techniques like pseudonymization and blockchain-enabled decentralized authentication), and robust governance protocols.

Specifically, the studies demonstrate that:

1. Persistent identifiers and metadata standards (e.g., HL7 FHIR, BIDS) enhance data discovery and interoperability.
2. Containerization and federated storage improve reproducibility, scalability, and resilience of infrastructure.
3. Privacy-preserving distributed data mining and policy-based access control support data security and compliance.
4. Community-driven standards and governance frameworks help integrate ethical considerations into infrastructure design.

Taken together, these contributions detail how FAIR principles are applied to build infrastructures that are both ethically responsible and resilient to technical challenges.

Paper search

Using your research question "How do FAIR data principles contribute to developing ethical and resilient infrastructure for AI research across scientific and industrial domains?", we searched across over 126 million academic papers from the Semantic Scholar corpus. We retrieved the 50 papers most relevant to the query.

Screening

We screened in sources that met these criteria:

- **FAIR Principles and AI:** Does the study examine FAIR (Findable, Accessible, Interoperable, Reusable) data principles implementation in the context of AI systems or infrastructure?
- **AI Infrastructure Focus:** Does the research address AI infrastructure development or deployment with explicit consideration of data management practices?
- **Ethical Considerations:** Does the study include analysis of ethical considerations in AI data management or FAIR principles implementation?

- **Implementation Evidence:** Does the study provide empirical evidence, practical implementation details, or case studies from scientific or industrial settings?
- **System Interoperability:** Does the research examine interoperability or integration aspects between AI systems and data repositories?
- **Infrastructure Resilience:** Does the study evaluate infrastructure resilience or sustainability aspects of AI systems implementing FAIR principles?
- **Review Currency:** If this is a systematic review or meta-analysis, was it published within the last 5 years? (Answer 'Yes' if this is not a systematic review/meta-analysis)
- **Evidence Quality:** Does the study present original empirical evidence, implementation data, or systematic analysis (rather than just opinion or commentary)?
- **Implementation Scope:** Does the research address both technical specifications AND practical/ethical implementation considerations?

We considered all screening questions together and made a holistic judgement about whether to screen in each paper.

Data extraction

We asked a large language model to extract each data column below from each paper. We gave the model the extraction instructions shown below for each column.

- **Domain and Research Focus:**

Identify the specific scientific or industrial domain addressed in the study. Look for explicit mentions of the research domain in the introduction, methods, or results sections. If multiple domains are discussed, list all of them. Examples might include materials science, AI research, photon science, etc.

Extraction guidelines:

- Extract the primary domain(s) mentioned
- If no clear domain is specified, write "Not specified"
- If multiple domains are equally emphasized, list all domains

- **FAIR Principles Implementation Approach:**

Describe how the study specifically implemented or addressed FAIR (Findable, Accessible, Interoperable, Reusable) data principles.

Extraction guidelines:

- Identify which specific FAIR principles were addressed
- Note any technological or methodological innovations used to implement FAIR principles
- Extract specific strategies or technologies used to enhance data management
- If no explicit FAIR implementation is described, write "No specific FAIR implementation detailed"

Look primarily in:

- Methods section
- Discussion of data infrastructure
- Technological framework descriptions
- **Computational Framework and Technologies:**

Identify and list the specific computational technologies, platforms, or infrastructures used in the study to support FAIR data principles or AI research.

Extraction guidelines:

- List all computational platforms, facilities, or systems mentioned
- Include specific technologies or software used
- Note any unique technological integrations
- If no specific technologies are detailed, write "No specific technologies described"

Look in:

- Methods section
- Technology description paragraphs
- Infrastructure discussion

- **Ethical Considerations and Resilience Strategies:**

Extract any explicit discussions of ethical considerations or strategies for ensuring data/AI infrastructure resilience.

Extraction guidelines:

- Identify any ethical frameworks or guidelines mentioned
- Note strategies for data protection, privacy, or responsible AI
- Capture any discussions of infrastructure robustness or resilience
- If no ethical considerations are discussed, write "No ethical considerations detailed"

Primary search areas:

- Ethics section (if present)
- Discussion of data management
- Methodological considerations

- **Key Findings and Implications for FAIR Data and AI Research:**

Summarize the primary findings related to FAIR data principles, AI research infrastructure, or data management strategies.

Extraction guidelines:

- Extract main research outcomes
- Capture any significant insights about FAIR data implementation
- Note implications for future research or practice
- If no clear findings are presented, write "No specific findings detailed"

Look in:

- Results section
- Discussion section
- Conclusion paragraphs

Results

Characteristics of Included Studies

Study	Study Focus	Domain Type (Scientific/Industrial)	FAIR Implementation Level	Infrastructure Components	Full text retrieved
Álvarez-Romero et al., 2022	Application of FAIR principles to health research data, federated machine learning, privacy-preserving data mining	Scientific (Health Research)	Full implementation: workflow, tools, federated learning, General Data Protection Regulation (GDPR) compliance	FAIR4Health workflow, Data Curation Tool, Data Privacy Tool, FAIR4Health Platform, Privacy-Preserving Distributed Data Mining (PPDDM) Agent, Health Level Seven Fast Healthcare Interoperability Resources (HL7 FHIR) Repository, Common Data Model	Yes

Study	Study Focus	Domain Type (Scientific/Industrial)	FAIR Implementation Level	Infrastructure Components	Full text retrieved
Ravi et al., 2022	FAIR principles for artificial intelligence models, unified computational framework, high energy diffraction microscopy	Scientific (Artificial Intelligence, Materials, Photon Science)	Full implementation: FAIR for artificial intelligence models, quantification, infrastructure integration	Advanced Photon Source, Materials Data Facility (MDF), Data and Learning Hub for Science (DLHub), funcX, Argonne Leadership Computing Facility (ALCF), ThetaGPU, SambaNova, Docker, Apptainer, Jupyter, Kubernetes	Yes
Duarte et al., 2022	FAIR artificial intelligence models in high energy physics, template and example	Scientific (High Energy Physics)	Full implementation: FAIR for artificial intelligence models, template, metadata, containerization	Scikit-learn, TensorFlow, PyTorch, Open Neural Network Exchange (ONNX), coo- iecutter4fair, Docker, GitHub, DLHub, Hugging Face, NVIDIA Triton Server, Apptainer	Yes

Study	Study Focus	Domain Type (Scientific/Industrial)	FAIR Implementation Level	Infrastructure Components	Full text retrieved
Mou et al., 2023	FAIR data in distributed machine learning, blockchain-based federated learning	Scientific (Artificial Intelligence, Machine Learning)	Full implementation: blockchain, decentralized identifiers, metadata, access control	Data Space, Blockchain Network, Smart Contracts, Docker, Ethereum Gateway, Hyperledger Fabric Gateway	Yes
Wilkinson et al., 2016	Introduction of FAIR Guiding Principles	Cross-domain (Life, Space Sciences)	Conceptual: foundational principles, examples of implementation	Dataverse, FAIRDOM, Investigation-Study-Assay (ISA), Open PHACTS, Worldwide Protein Data Bank (wwPDB), UniProt, Data Citation Implementation Group	Yes
Bade, 2025	Ethical and sustainable artificial intelligence infrastructure, sectoral applications	Cross-sectoral (Finance, Healthcare, Retail)	We didn't find mention of a specific FAIR implementation in the full text or abstract	We didn't find mention of specific technologies in the full text or abstract	No

Study	Study Focus	Domain Type (Scientific/Industrial)	FAIR Implementation Level	Infrastructure Components	Full text retrieved
Aggour et al., 2024	Semantics-enabled data federation for FAIR data in materials science	Scientific/Industrial (Materials, Aerospace, Power)	Full implementation: federated storage, knowledge graphs, FAIR services	Systems and Architecture Description Language (SADL), Semantic Toolkit (SemTK), Python functions, Amazon Web Services (AWS) Gov-Cloud	Yes
Reer et al., 2023	FAIR neuroscientific data sharing, legal frameworks, metadata standards	Scientific (Neuroscience, Artificial Intelligence)	Full implementation: metadata standards, legal/ethical compliance	Brain Imaging Data Structure (BIDS), Hierarchical Event Descriptors (HED), Neuroimaging Data Model (NIDM), openMINDS, BIDS-Matlab, PyBIDS, DataLad, MetaLad, DataCat, PyNIDM, Vantage6, OpenMined	Yes

Study	Study Focus	Domain Type (Scientific/Industrial)	FAIR Implementation Level	Infrastructure Components	Full text retrieved
Bailo et al., 2022	FAIR data management in solid Earth science (European Plate Observing System, EPOS)	Scientific (Solid Earth Science)	Full implementation: multi-tier architecture, governance, metadata	EPOS Data Portal, Graphical User Interface (GUI), Integrated Core Services Central (ICS-C), Integrated Core Services Distributed (ICS-D), Thematic Core Services (TCS), SWIRRL-API, Kubernetes, JupyterLab, Enlighten-web	Yes
Govarts et al., 2022	Management of personal data in environment and health research, GDPR/FAIR balance	Scientific (Environment & Health)	We didn't find mention of a specific FAIR implementation in the full text or abstract (abstract only)	Federated data management, machine learning, advanced search, ontologies, European Open Science Cloud (EOSC), European Infrastructure for Research in Environmental Epidemiology (EIRENE)	No

Domain Type:

- Seven studies focused on scientific domains (health, artificial intelligence, materials, neuroscience, solid earth, environment and health).
- One study addressed a mixed scientific/industrial domain (materials, aerospace, power).
- One study was cross-domain (life and space sciences).

- One study was cross-sectoral (finance, healthcare, retail).

FAIR Implementation Level:

- Seven studies described full implementation of FAIR principles.
- One study provided only a conceptual discussion of FAIR principles.
- Two studies did not detail a specific FAIR implementation.

Infrastructure Components:

- Containerization technologies (Docker, Apptainer) were mentioned in four studies.
- Data and compute platforms (such as Data and Learning Hub for Science, Materials Data Facility, Advanced Photon Source) were found in three studies.
- Metadata standards and ontologies (e.g., Health Level Seven Fast Healthcare Interoperability Resources, Brain Imaging Data Structure, Hierarchical Event Descriptors, Neuroimaging Data Model, openMINDS, Investigation-Study-Assay, Data Citation) were mentioned in four studies.
- Federated learning or federated data management was described in three studies.
- Blockchain and smart contract infrastructure was found in one study.
- Semantic and knowledge graph tools were found in one study.
- Cloud infrastructure (Amazon Web Services Gov-Cloud, European Open Science Cloud, European Infrastructure for Research in Environmental Epidemiology) was mentioned in two studies.
- Machine learning frameworks (Scikit-learn, TensorFlow, PyTorch, Open Neural Network Exchange) were found in one study.
- Code and data sharing platforms (GitHub, Hugging Face) were found in one study.
- Privacy-preserving or federated computation tools (Privacy-Preserving Distributed Data Mining Agent, Vantage6, OpenMined) were found in two studies.
- User interface and interactive tools (Jupyter, JupyterLab, Graphical User Interface, Enlighten-web) were found in three studies.
- Orchestration tools (Kubernetes) were found in two studies.
- Data portals were found in two studies.
- We didn't find mention of specific technologies in one study.

Thematic Analysis

FAIR Principles as Infrastructure Foundation

FAIR Component	Infrastructure Impact	Implementation Challenges	Success Factors
Findable	Persistent identifiers (Digital Object Identifiers, Decentralized Identifiers), metadata catalogues, federated storage enable data/model discovery across platforms	Heterogeneous data sources, lack of standardization, metadata quality	Adoption of community standards, rich metadata, knowledge graphs

FAIR Component	Infrastructure Impact	Implementation Challenges	Success Factors
Accessible	Open application programming interfaces (APIs), distributed/federated access, privacy-preserving data mining, blockchain gateways	Legal/ethical restrictions (General Data Protection Regulation), authentication/authorization complexity	Privacy-preserving computation, decentralized authentication, policy-based access
Interoperable	Standardized metadata (Brain Imaging Data Structure, Data Catalog Vocabulary, Health Level Seven Fast Healthcare Interoperability Resources), knowledge graphs, containerization, APIs	Domain-specific vocabularies, legacy systems, cross-platform compatibility	Use of ontologies, open standards, containerized environments
Reusable	Rich metadata, provenance tracking, containerized models, open-source tools, analytic chaining	Data quality, legal reuse restrictions, lack of incentives	Community-driven standards, automated documentation, governance frameworks

Infrastructure Impact:

- Containerization or containerized models/environments were mentioned for three FAIR components (Interoperable, Reusable, and indirectly for Interoperable via APIs).
- Federated or distributed access/storage was mentioned for two components (Findable, Accessible).
- Open APIs were mentioned for two components (Accessible, Interoperable).
- Knowledge graphs were mentioned for two components (Findable, Interoperable).
- Rich metadata was mentioned for two components (Findable, Reusable).
- Privacy-preserving data mining or computation was mentioned for two components (Accessible, Reusable).
- Persistent identifiers, metadata catalogues, blockchain gateways, standardized metadata, provenance tracking, open-source tools, and analytic chaining were each mentioned for one component.

Implementation Challenges:

- Legal/ethical restrictions (including General Data Protection Regulation and legal reuse) were mentioned for two components (Accessible, Reusable).
- Lack of standardization or domain-specific vocabularies was mentioned for two components (Findable, Interoperable).
- Other challenges (each mentioned for one component) included: heterogeneous data sources, metadata

quality, authentication/authorization complexity, legacy systems, cross-platform compatibility, data quality, and lack of incentives.

Success Factors:

- Adoption of community standards or community-driven standards was mentioned for two components (Findable, Reusable).
 - Other success factors (each mentioned for one component) included: rich metadata, knowledge graphs, privacy-preserving computation, decentralized authentication, policy-based access, use of ontologies, open standards, containerized environments, automated documentation, and governance frameworks.
 - We didn't find any infrastructure impacts, challenges, or success factors that were reported for all four FAIR components. Most categories were specific to one or two components.
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Ethical Framework Integration

- Health and neuroscience studies : Emphasized General Data Protection Regulation compliance and privacy-preserving computation, with explicit strategies such as privacy-preserving distributed data mining, pseudonymization, and informed consent.
 - Blockchain and decentralized authentication : Used for data protection in distributed machine learning contexts.
 - Governance and policy frameworks : Highlighted as essential for ethical data management, especially in large-scale scientific infrastructures.
 - Technical focus : Some studies focused primarily on technical aspects, with ethical implications implicit in adherence to FAIR principles.
 - Abstract-only studies : One study foregrounded ethical artificial intelligence infrastructure, including transparency, fairness, and sustainability, but lacked implementation detail.
 - Resilience and robustness : Often discussed in the context of governance and technical architecture, rather than as explicit ethical constructs.
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Cross-Domain Infrastructure Resilience

- Federated and distributed architectures : Cited as mechanisms for resilience, supporting scalability and adaptability.
 - Containerization : Used to enhance reproducibility and portability across platforms.
 - Robust governance : Identified as a factor in sustaining resilient infrastructure.
 - Blockchain and decentralized identifiers : Highlighted for auditability, provenance, and trust.
 - Knowledge graphs and multi-tier architectures : Supported scalability and adaptability.
 - Empirical evaluation : Explicit evaluation of infrastructure resilience (e.g., in the face of failures or attacks) was limited; most evidence was architectural or conceptual.
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Implementation Challenges and Solutions

Challenge Type	Domain Impact	Mitigation Strategy	Outcome
Legal/Ethical (General Data Protection Regulation, privacy)	Health, neuroscience, environment	Privacy-preserving computation, pseudonymization, federated learning, decentralized authentication	Secure, compliant data sharing; ongoing challenges with cross-jurisdictional data
Metadata/Standardization	All domains	Adoption of community standards (Brain Imaging Data Structure, Health Level Seven Fast Healthcare Interoperability Resources, Data Catalog Vocabulary), knowledge graphs, minimal metadata schemes	Improved interoperability and reusability; challenges with legacy data
Technical Integration	All domains	Containerization, application programming interfaces, federated storage, multi-tier architectures	Enhanced scalability, reproducibility, and cross-platform compatibility
Governance/Sustainability	Earth science, materials, cross-domain	Policy frameworks, governance boards, community engagement	Sustainable, community-driven infrastructure; slow adoption of some FAIR components

- Four distinct challenge types were addressed across the studies: Legal/Ethical, Metadata/Standardization, Technical Integration, and Governance/Sustainability.
- Mitigation strategies were unique to each challenge type, including privacy-preserving computation, pseudonymization, federated learning, decentralized authentication, adoption of community standards, knowledge graphs, minimal metadata schemes, containerization, application programming interfaces, federated storage, multi-tier architectures, policy frameworks, governance boards, and community engagement.
- Reported outcomes were also unique to each challenge type, including secure and compliant data sharing, improved interoperability and reusability, enhanced scalability and reproducibility, and sustainable, community-driven infrastructure. Ongoing challenges were noted for cross-jurisdictional data, legacy data, and slow adoption of some FAIR components.
- We did not find any mitigation strategy or outcome that was reported in more than one study.

References

Aaron Reer, Andreas Wiebe, Xu Wang, and J. Rieger. “FAIR Human Neuroscientific Data Sharing to Advance AI Driven Research and Applications: Legal Frameworks and Missing Metadata Standards.”

- Frontiers in Genetics*, 2023.
- Celia Álvarez-Romero, A. Martínez-García, A. Sinaci, M. Gencturk, Eva Méndez, Tony Hernández-Pérez, Rosa Liperoti, et al. “FAIR4Health: Findable, Accessible, Interoperable and Reusable Data to Foster Health Research.” *Open Research Europe*, 2022.
- D. Bailo, K. Jeffery, K. Atakan, L. Trani, R. Paciello, Valerio Vinciarelli, J. Michałek, and A. Spinuso. “Data Integration and FAIR Data Management in Solid Earth Science.” *Annals of Geophysics*, 2022.
- Govarts Eva, Gilles Liese, Bopp Stephanie, Holub Petr, Matalonga Leslie, Vermeulen Roel, Vrijheid Martine, et al. “Position Paper on Management of Personal Data in Environment and Health Research in Europe.” *Environment International*, 2022.
- Javier Mauricio Duarte, Haoyang Li, Avik Roy, Ruike Zhu, E. Huerta, Daniel Diaz, P. Harris, et al. “FAIR AI Models in High Energy Physics.” *Machine Learning: Science and Technology*, 2022.
- K. Aggour, Vijay S. Kumar, Vipul K. Gupta, Alfredo Gabaldon, P. Cuddihy, and Varish Mulwad. “Semantics-Enabled Data Federation: Bringing Materials Scientists Closer to FAIR Data.” *Integrating Materials and Manufacturing Innovation*, 2024.
- Mark D. Wilkinson, M. Dumontier, I. J. Aalbersberg, Gabrielle Appleton, M. Axton, A. Baak, N. Blomberg, et al. “The FAIR Guiding Principles for Scientific Data Management and Stewardship.” *Scientific Data*, 2016.
- Nikil Ravi, Pranshu Chaturvedi, E. Huerta, Zhengchun Liu, Ryan Chard, Aristana Scourtas, K. J. Schmidt, K. Chard, B. Blaiszik, and Ian T. Foster. “FAIR Principles for AI Models with a Practical Application for Accelerated High Energy Diffraction Microscopy.” *Scientific Data*, 2022.
- Shivakrishna Bade. “AI Infrastructure and Its Societal Impact: Building Ethical and Sustainable Systems.” *International Journal for Sciences and Technology*, 2025.
- Yongli Mou, Fengyang Guo, Wei Lu, Yongzhao Li, O. Beyan, Thomas Rose, S. Dustdar, and Stefan Decker. “Towards FAIR Data in Distributed Machine Learning Systems.” *Global Communications Conference*, 2023.